Programme : **Diploma in Cloud Computing and Big Data**	
Course Code : **5279A**	Course Title: **Introduction to Microservices Lab**
Semester : **5**	Credits: **1.5**
Course Category: **Programme Elective course**	
Periods per week: **3 (L: 0 T: 0 P: 3)**	Periods per semester: **45**

Course Objectives:

- To strengthen basic concepts of HTML and Javascript
- To Learn to install and use Docker on Linux
- Learn to make use of MVC framework to develop web application
- Be able to deploy web application on docker

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Python	3279	Programming in Python Lab	3
Database management concepts	3275	Database Lab	3
HTML & Javascript	4277	Web Application Development Lab	4
Docker	4279	Advanced Operating System	4

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Implement a simple ecommerce site using HTML, CSS and JavaScript	8	Applying
CO2	Demonstrate creation and deployment of dockers	10	Applying
CO3	Practice microservice API using Model View controller framework	10	Applying

CO4	Interact with a microservices system built using Spring Boot	10	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			2			
CO2	3			2			
CO3	3			2			
CO4	3			2			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	**Implement a simple ecommerce site using HTML, CSS and JavaScript**		
M1.01	Implement simple HTML pages using Basic HTML tags	2	Applying
M1.02	Demonstrate creation of web pages using Inline, Embedded and External Style Sheets	2	Applying
M1.03	Use JavaScript to make web pages more dynamic	2	Applying
M1.04	Integrate various web pages to construct an ecommerce site	2	Applying
Contents Revisit Website design and development concepts. Contents- Basic HTML tags, lists, hyperlinks, images, data table. CSS properties – font, color, background, list, link, text, border in web pages. JavaScript- Built-in function and model Event Handling. Web site optimization techniques- best practices- overview.			

CO2	Demonstrate creation and deployment of dockers		
M2.01	Install docker engine and run a hello-world docker image	2	Applying
M2.02	Demonstrate the use of External (Pre-Built) Docker Images	1	Applying
M2.03	Illustrate the installation of software packages on Docker	2	Applying
M2.04	Demonstrate the working of Docker Volumes and Networks.	5	Applying
Contents Installation of software Packages- Container- Packages like PHP, MySQL and git Docker Volumes and Networks : Mounting host machine harddrive, connecting to host machine network.			
	Lab Test – I	3	
CO3	**Practice microservice API using Model View Controller framework.**		
M3.01	Illustrate the Installation of MVC framework	1	Applying
M3.02	Deploy a web application using MVC framework	2	Applying
M3.03	Practice connecting a web application to MySQL	4	Applying
M3.04	Deploy the web application in docker	2	Applying
M3.05	Develop a Microservice API with Django Rest Framework (DRF)	3	Applying
Contents Install MVC framework like Django. MySQL- Database connectivity- Create, read, update, delete Django Rest Framework (DRF) - install django Rest framework, DRF Serializers, API Requests through GET and POST.			

CO4	**Interact with a microservices system built using Spring Boot**		
M4.01	Interact with a complete microservice architecture implemented using spring boot	8	Understand
	Lab Test – II	3	
Contents Familiarize and identify the components of an already implemented microservices system implemented using spring boot.			

Text / Reference

T/R	Book Title/Author
T1	Beginning Django E-Commerce, JIM MCGAW, Apress
T2	Practical Docker with Python Build, Release and Distribute your Python App with Docker, Sathyajith Bhat ,Apress
R3	"Hands-On Microservices with Spring Boot and Spring Cloud: Build and deploy Java microservices using Spring Cloud, Istio, and Kubernetes", Magnus Larsson , Packt

Online Resources

Sl.No	Website Link
1	https://www.youtube.com/watch?v=ddrucr_aAzA
2	https://www.youtube.com/watch?v=0iB5IPoTDts
3	https://hub.docker.com/